Report

Prussian Blue: Artists' Pigment and Chemists' Sponge

by Mike Ware

In 1981 this *Journal* published an article by Andreas Ludi *(1)* entitled: "Prussian Blue—An Inorganic Evergreen"—a neat oxymoron![1] Since then, Prussian blue has often been in the news. Ferric ferrocyanide, or iron(III) hexacyanoferrate(II) as we chemists must now call it, is famous for two quite different chemical properties: its color, due to the mixed oxidation states of iron; and its zeolytic character—a consequence of the open structure of its crystal lattice. (Zeolites belong to a complex family of porous alkali metal aluminosilicate minerals, but the adjective is now applied to any substance capable of taking up and trapping other molecules or ions in the cavities of its lattice, and acting as a molecular sieve or ion exchanger.) Prussian blue's capability for hosting small molecules and ions explains why it took chemists 200 years to understand the contradictory analytical data presented by the several varieties of this substance.

Blue History

Also called Paris blue or Milori blue, the politically-correct[2] name for Prussian blue in the German-speaking world is *Berliner blau*—after the place of its first, accidental, preparation in 1704. The story begins then, with the artists' colormaker, Heinrich Diesbach of Berlin, trying to prepare a crimson lake. ("Lakes" are artists' colors formed of precipitates with adsorbed dyes, and crimson lake is made from alum plus potash (potassium carbonate) precipitating aluminium hydroxide, to which is added the dye cochineal, extracted from crushed beetles.) Diesbach lacked the necessary potash, so he purchased some from a notorious local alchemist, Johann Konrad Dippel, who came from Castle Frankenstein near Darmstadt. (Having failed to make gold, Dippel launched a medicinal "animal oil" upon an unsuspecting public in 1700.) Imagine Diesbach's surprise when the color he obtained was not crimson, but blue. It turned out that Dippel's potash had been contaminated with his infamous "animal oil". This elixir was a malodorous distillate of animal carcasses—blood, bones, and offal. At the time it was hailed as a panacea—presumably sustained by a widely-held belief that anything so obnoxious must be beneficial. We now know it consisted of a mixture of nitrogenous organic bases such as pyrrole, and several alkyl cyanides, arising from the thermal degradation of molecules containing C–N bonds, such as hemoglobin. By chance, it provided the essential, but then unknown, ingredient—cyanide—to enable Diesbach's serendipitous discovery of Prussian blue. All painters since then have reason to be grateful to the unscrupulous alchemist Dippel. Prussian blue does not occur in nature, so has been claimed as the first synthetic pigment, but this title must now be ceded to Egyptian blue: a calcium copper(II) silicate, whose recipe was lost in the Roman era and only recently rediscovered *(2)*. Prussian blue may still qualify, however, as the first synthetic coordination compound.

Although the organic component needed for Prussian blue had been identified, its preparation was kept secret until 1724 when John Woodward finally published, in Latin, a method that called for dried ox-blood as the starting material *(3)*. From the thermal degradation of this iron-containing nitrogenous organic matter by potash, the initial product is the iron complex salt we now call potassium ferrocyanide, but its origins are evident in the antique German name for this substance: *blutlaugensaltz*, "blood-residue salt". It was then known as "prussiate of potash" and was most conveniently made by the destruction of Prussian blue by alkalies, as discovered by Pierre-Joseph Macquer in 1752. So the accidental discovery of Prussian blue opened a doorway upon a whole new field of chemistry—of the cyanide group, which consequently takes its name from the Greek word for dark blue, *kyaneos,* although most cyanides are colorless! It is remarkable that all this chemistry took place *before* nitrogen had even been identified as an element, in 1772 by Daniel Rutherford; nitrogen was also recognized around this time by Scheele, Cavendish, and Priestley as "dephlogisticated" or "burnt" air. Scheele in 1782 was the first to isolate hydrogen cyanide—he called it "prussic acid"—from Prussian blue by heating it with concentrated sulfuric acid, and he was lucky to survive his first encounter with this speedy, lethal poison!

By 1814 Joseph Louis Gay-Lussac had prepared the parent molecule cyanogen, $(CN)_2$, (which he called in French *cyanogène;* from the Greek *kyaneos-gennao* meaning "blue-generating"). With the realization that the cyanide radical was composed just of carbon tightly combined with nitrogen, purely inorganic methods were proposed for synthesizing potassium ferrocyanide: from charcoal, potash, iron filings, and atmospheric nitrogen. This welcome alternative must have provided a merciful relief from the "fetid vapors" of the organic route! Prussian blue could then easily be prepared by reaction of ferric salts with this "prussiate of potash" (Figure 1). We have all at some time been visually astonished by this simple reaction: two unremarkable aqueous solutions, one pale yellow, the other colorless, when mixed turn an inky blue-black:[3]

$$Fe^{3+}(aq) + K^{+}(aq) + [Fe^{II}(CN)_6]^{4-}(aq) \longrightarrow KFe^{III}[Fe^{II}(CN)_6](s)$$

Figure 1. Formation of Prussian blue. Video of this reaction is available in the online material. [Photo and video by James Maynard and J. W. Moore.]

Figure 2. Painting of a student and teacher observing the Prussian blue reaction. Alfred Bader, owner of this painting, has named it "The Preparation of Prussian Blue". [Courtesy of Alfred Bader.]

Performance of this dramatic reaction was part of the stock-in-trade for the public lecture-demonstrations of celebrated 19th-century chemists such as Michael Faraday, who is seen in the painting shown in Figure 2 as a youth, looking on enthralled as William Thomas Brande performs the reaction *(4).*

Crystal Structure of the Blues

The composition of iron(III) hexacyanoferrate(II) can vary between "insoluble" Prussian blue (IPB for short): $Fe^{III}{}_4[Fe^{II}(CN)_6]_3 \cdot xH_2O$ where x = 14–16, and so-called "soluble" Prussian blue (SPB): $M^IFe^{III}[Fe^{II}(CN)_6] \cdot yH_2O$ where y = 1–5. M is a monovalent cation such as K^+ or NH_4^+. (For ease of comprehending the formulas, the oxidation states are indicated in superscripted Roman numbers.) "Soluble" Prussian blue belies its name; it is actually highly insoluble, but gives the appearance of forming a solution in water because it is easily peptized as a blue colloidal sol, which passes through a filter and does not sediment.

Apart from its color, the next clue that there is something unusual about Prussian blue is indicated by its density: ca. 1.8 g cm^{-3}. Why is it so very light compared with similar iron salts, whose densities are >3 g cm^{-3}? The answer was found by Ludi and coworkers, who successfully grew single crystals of Prussian blue and determined its crystal structure by X-ray diffraction *(5).* As shown at the left in Figure 3 *(6)* the iron atoms define a simple cubic lattice, alternating Fe(II) and Fe(III), and are linked by linear cyanide groups, with their carbon atoms coordinated to the Fe(II). This open scaffolding leaves large cavities within, and tunnels running in three directions throughout the roomy lattice. The centers of the cubic cells can be occupied by ions or molecules up to 182 pm in radius. In SPB, half of these cavities are usually occupied by K^+, and the "ideal" structure fits the formula above (center, Figure 3). In IPB however, the lattice must be imperfect in order to balance the charges and agree with the formula: since the ratio of Fe(III):Fe(II) = 4:3, a quarter of the iron(II), as $[Fe(CN)_6]^{4-}$, must be absent; the vacant sites are occupied by water molecules instead, and the coordination sphere around the Fe(III) centers contains water, with the average composition $[Fe(NC)_{4.5}(OH_2)_{1.5}]$ (shown at the right in Figure 3).

Trapping Toxic Cations

In the mid-20th century, the heavy metal thallium, at the foot of Group III, enjoyed a spell of notoriety as a subtle poison: its unusual symptoms could be mistaken for diseases such as encephalitis or epilepsy, but incidentally the victim's hair dropped out, so sudden alopecia offered a clue *(7).* (Its use as a depilatory, however, declined in the 1930s.) Thallium(I) sulfate is colorless, odorless, tasteless, and sufficiently water-soluble to be administered easily to an unsuspecting victim. The symptoms do not appear for several days, but a dose of a gram or so is usually fatal because the body cannot get rid of the element, which mimics potassium but blocks its vital enzymic action, especially

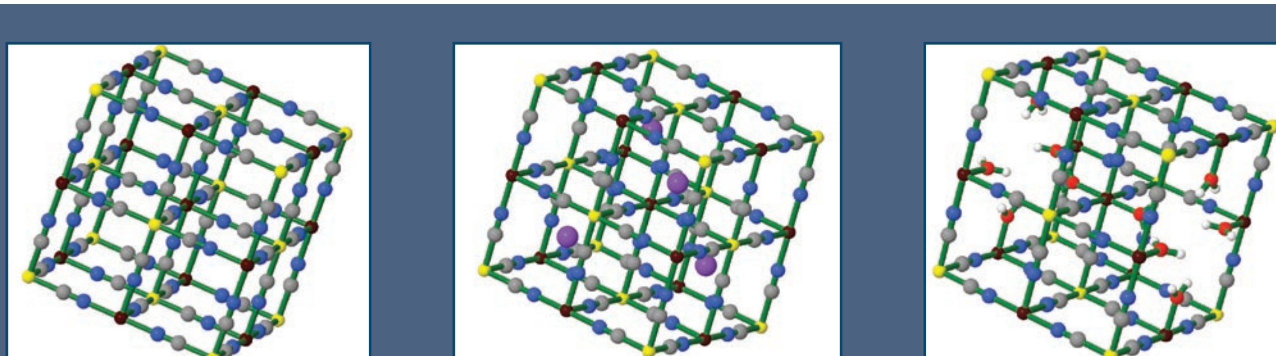

Figure 3. At left, simple cubic lattice of Prussian blue. Fe^{II} yellow; Fe^{III} red; C gray; N blue. Center, potassium ions (purple) occupy the centers of half of the cubic cells. At right, in $Fe_4[Fe(CN_6)]_3 \cdot xH_2O$ there are fewer Fe^{II} ions than Fe^{III} ions, which leaves some Fe^{II} sites vacant; the vacant sites are occupied by water molecules. Dynamic Jmol structures are available in the online material. [Structures courtesy of Xavier Prat-Resina.]

in the brain. Thallium was used quite extensively by the secret police of the late Saddam Hussein's regime in Iraq for disposing of senior political dissidents *(8)*. A useful trick was to administer the dose just before the victim was allowed to leave the country, so he died abroad. The substance was readily available in the Middle East, where it was commonly employed to destroy rats and cockroaches, so accidental poisoning was also common.

Fortunately—if the cause of the poisoning is correctly diagnosed—there is an antidote that has saved many lives: Prussian blue. The thallium(I) ion, Tl^+, has a (6-coordinate) radius of 164 pm, and can enter the Prussian blue lattice in place of potassium, radius 152 pm. There it becomes trapped, and can be safely excreted by the body. Prussian blue itself is non-toxic, unreactive, and tasteless, and can be safely eaten in gram quantities without harmful effect. It was therefore put forward by German pharmacologist, Horst Heydlauf of Karlsruhe, as the preferred antidote for thallium poisoning, and its efficacy has since been amply confirmed *(9)*.

The Group I metal, cesium, also possesses a large, singly-charged cation, with a radius of 181 pm, which can be absorbed and "imprisoned" by Prussian blue. Unlike thallium(I), cesium(I) is not particularly toxic chemically, but a radioisotope of this element does cause serious concern. The fission product, Cs^{137}, formed in nuclear reactors and nuclear explosions, is a powerful beta- and gamma-emitter with a half-life of 30 years; it remains in the body for 110 days or more, and is therefore a potent radiotoxin.

In April 1986, there was a disastrous nuclear reactor explosion at Chernobyl that emitted a radioactive cloud containing Cs^{137}, among other radioisotopes. The cloud drifted westward across Europe, depositing radioactive rain. In the northwest United Kingdom, the upland pastures of Cumbria and North Wales were particularly vulnerable. Cesium is taken up by the grass that is grazed by sheep and so can pass into the human food chain. Tests were conducted to see whether this contamination could be countered by spreading Prussian blue on the pastures—so turning the green hills of England to navy blue!—but it was found more effective to feed the Prussian blue directly to the affected sheep, and thus protect our Sunday lunch *(10)*. The effects of Chernobyl lingered on. In 1992, it was discovered that migrant herds of reindeer in Norway had also absorbed Cs^{137} from this disaster, because they graze on lichens and are particularly partial to mushrooms, both of which concentrate the cesium. Prussian blue was therefore added to the salt licks that these nomadic animals are so fond of, with beneficial effects *(11)*. Prussian blue continues to be used to detoxify the livestock in the Chernobyl-contaminated regions of Belarus, Russia, and Ukraine, and has saved their agricultural industries of milk and meat production from enormous losses.

Cs^{137} is also used in radiotherapy to treat certain cancers, so there are substantial quantities of this radioisotope in specialist medical facilities worldwide. Acknowledging the risk that these stocks could be abused by terrorists to make a radiological "dirty bomb", the Federal Drug Administration in the U.S. has recently licensed the use of Prussian blue as a safe antidote to Cs^{137} contamination in humans *(12)*.

Table 1. Iron/Cyanide Complexes

Components	$[Fe^{III}(CN)_6]^{3-}$	$[Fe^{II}(CN)_6]^{4-}$
Fe^{3+}	**Ferric ferricyanide** Prussian yellow or Berlin yellow Powerful oxidant, soluble being reduced to Prussian blue	**Ferric ferrocyanide** Prussian blue or Berlin blue The redox-stable pigment hydrolyzed by alkali
Fe^{2+}	**Ferrous ferricyanide** Highly unstable to electron transfer From Fe(II) to Fe(III) giving **Ferric ferrocyanide** Turnbull's blue = Prussian blue	**Ferrous ferrocyanide** Berlin white, or Everitt's salt or Williamson's salt. Colorless. Easily oxidized by air, etc. to Prussian blue

Redox Family Relations

The color of Prussian blue is due to a broad absorption band peaking around 680 nm, which corresponds to the energy of transferring an electron from Fe(II) to Fe(III) and is therefore a direct consequence of the mixed valence character of the substance. Prussian blue may be oxidized to Prussian yellow, containing all iron(III), $Fe^{III}[Fe^{III}(CN)_6]$, or reduced to Prussian white, containing all iron(II), $Fe^{II}[Fe^{II}(CN)_6]^{2-}$, that has several alternative names: Berlin white, Everitt's salt, or Williamson's salt. This redox chemistry is summarized in Table 1 above. The electron transfers take place without changing the essential lattice structure already described, although they do require a flow of cations in or out of the lattice to balance the change in overall electric charge.

The useful artists' color called Turnbull's blue was first produced in the 18th century by the color manufacturers, Messrs. J. M. & W. Turnbull, in East Lothian, Scotland. It is made by adding ferrous salts to ferricyanides and was long thought to be ferrous ferricyanide—chemically different from Prussian blue. However, Mössbauer spectroscopy in the 1970s showed the two pigments to be essentially the same substance, ferric ferrocyanide, despite the differing methods of manufacture:

$$Fe^{2+}(aq) + [Fe^{III}(CN)_6]^{3-}(aq) \longrightarrow Fe^{III}[Fe^{II}(CN)_6]^{-}(s)$$

The preparation of true ferrous ferricyanide has been reported *(13)*, but this claim has been questioned *(14)*. It is certainly a highly unstable substance that instantly reverts to ferric ferrocyanide.

The changes of color according to the redox state of the iron qualifies Prussian blue as an *electrochromic* material, because these changes can be brought about by applying an electric potential across an electrode bearing a film of Prussian blue. It therefore has great commercial application as a digital display in devices such

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Figure 4. Ultramarine was used in "Maestà", (Madonna with Child with 20 Angels and 19 Saints) by Duccio di Buoninsegna. It is tempera on wood. [Image from Wikimedia Commons, *http://en.wikipedia.org/wiki/Image:Duccio_Maest%C3%A0.jpg;* source *http://gallery.euroweb.hu/html/d/duccio/buoninse/maesta/maest_01.html* (both sites accessed Feb 2008).]

as portable calculators. The color changes—yellow to blue to colorless—may be prettily illustrated by an experiment using cyclic voltammetry, as described in this *Journal (15).*

Beautiful but Vulnerable Blue

Until Diesbach's discovery, blue was a problematic color for artists: there was no entirely satisfactory pigment. The plant dyestuff indigo (the *woad* of ancient Britons) was rather prone to fading. The permanent pigment ultramarine, made of the blue semi-precious mineral *lapis lazuli,* was mined only in Afghanistan and was exceedingly expensive. Other blues, such as smalt (a cobalt silicate glass) and azurite (basic copper(II) carbonate, $2Cu(CO_3)\cdot Cu(OH)_2$, also had their disadvantages. Ultramarine was so costly that patrons limited its use in their commissioned paintings: the preciousness of the pigment was reflected in its reservation for sacred subjects, such as the robes of religious figures (Figure 4) *(16).* The arrival of Prussian blue transformed artists' attitudes to the blue on their palettes; we can see an instance in the sudden outburst of generous application of the color in the work of the French Impressionist School.

Prussian blue as a pigment does have two disadvantages: First, it is vulnerable to alkaline hydrolysis, which completely destroys the blue color by converting Prussian blue to iron(III) hydroxide and ferrocyanide:

$$\mathrm{Fe^{III}[Fe^{II}(CN)_6]^-(aq) + 3OH^-(aq) \longrightarrow Fe(OH)_3(s) + [Fe^{II}(CN)_6]^{4-}(aq)}$$

Therefore the pigment cannot be used in fresco painting, where lime is employed. Second, Prussian blue sometimes shows a tendency to fade in the light, but this is a variable phenomenon. Because the light-fading is essentially a photoreduction to Prussian white, its occurrence depends on the presence of some impurity that can be correspondingly oxidized to supply the electrons:

$$\mathrm{Fe^{III}[Fe^{II}(CN)_6]^-(s) + e^- \xrightarrow{h\nu} Fe^{II}[Fe^{II}(CN)_6]^{2-}(s)}$$

Moreover, this is a wholly reversible reaction if the pigment has access to the air, because oxygen will slowly reoxidize Prussian white to blue:

$$\mathrm{4Fe^{II}[Fe^{II}(CN)_6]^{2-}(s) + O_2(aq) + 4H^+(aq) \longrightarrow 4Fe^{III}[Fe^{II}(CN)_6]^-(s) + 2H_2O(l)}$$

Pure Prussian blue pigment does not fade and is graded as very light fast; the circumstances of its fading when mixed have been extensively studied at the National Gallery, London *(17).*

Photochemical Blues

The blueprint process, otherwise known as the cyanotype, has been previously described in this *Journal (18)* and elsewhere *(19),* so only a brief summary is needed here. It was invented accidentally by Sir John Herschel in 1842, when he was trying to do something rather different. This was only three years after Henry Talbot and Louis Daguerre had separately announced their independent discoveries of photography in silver. Herschel was trying to devise a process of color photography, and he found that some organic dyes extracted from the flowers of his garden could be directly bleached by sunlight, and thus form the basis for a positive-working photographic process, which he called his "anthotype" process. Exposures were excessively long, however, so Herschel hoped that more sensitive bleaching might also be achieved with certain highly-colored inorganic substances.

A young chemical colleague in the Royal Society, Alfred Smee, provided Herschel with a sample of the bright red complex salt, potassium ferricyanide, which he had prepared by electrolytic oxidation of the ferrocyanide. The ferricyanide was then a rare compound because it was much more difficult to obtain in a pure state than the common ferrocyanide. Herschel coated paper with a solution of Smee's potassium ferricyanide, (in those days it was called "ferrosesquicyanuret of potash") and exposed it to the brilliant summer sun of 1842. Once again, the history of serendipitous discoveries repeated itself: Herschel was astonished to see that, far from bleaching out the red color, the sunlight turned it dark blue. Herschel instantly recognized that Prussian blue had been formed and used the process to photocopy engravings (Figure 5) *(20).* He was soon able to shorten

Figure 5. Sir John Herschel, cyanotype (1842) of an engraving of Mrs. Leicester Stanhope, by Charles Rolls (1836). [Used with permission of Photography Collection, Harry Ransom Humanities Research Center, The University of Texas at Austin.]

the lengthy exposures greatly when Smee also drew his attention to the "new" substance, ammonium ferric citrate, which was beginning to appear on the pharmacists' shelves as a popular iron tonic (it is still used today!) and a remedy for gastric complaints; Herschel discovered this to be even more light-sensitive.

The photosensitive coating for Herschel's cyanotype paper therefore consists of a mixture of ammonium iron(III) citrate and potassium ferricyanide. We now know that the action of near-UV and blue light causes a photochemical redox reaction in the iron complex, whereby the iron(III) is reduced to iron(II) and the citrate is oxidized initially to acetone dicarboxylic acid *(21)*.

$$2\,Fe^{3+}(aq) + C(OH)COOH(CH_2COOH)_2(aq) \xrightarrow{h\nu} 2\,Fe^{2+}(aq) + CO(CH_2COOH)_2(aq) + CO_2(g) + 2\,H^+(aq)$$

The iron(II) then couples with the ferricyanide to precipitate Prussian blue in a density proportional to the quantity of photons absorbed:

$$Fe^{2+}(aq) + \left[Fe^{III}(CN)_6\right]^{3-}(aq) \longrightarrow Fe^{III}\left[Fe^{II}(CN)_6\right]^{-}(s)$$

Excess unreacted chemicals are simply washed out with a water bath, leaving a tonally graduated, negative-working photograph in Prussian blue. Opaque objects just yield silhouettes of their outlines—called photograms—such as the cyanotype impressions of seaweeds and plants made by Anna Atkins, who was the first to use the process for botanical illustration (Figure 6) *(22)*. Photographic negatives, however, when contact-printed yield fully nuanced positive prints. To achieve a sufficient throughput of light for this rather insensitive process, which has a quantum yield of less than one, contact printing is generally necessary; attempts to employ the process in the camera are disadvantageous, demanding very long exposures and wide lens apertures.

The blueprint was the first commercial reprographic process, widely used in drawing offices for plan copying from the 1870s until the 1960s, when it was superseded by electrophotographic methods. Cyanotype was also used for proofing photographic negatives; some artists even acquired a taste for the unremitting and rather strident blue, and used it as an expressive medium in its own right. It is still popular today for its cheapness, low toxicity, and ease of processing, since it requires nothing but water; it provides an excellent hands-on introduction

Figure 6. Two botanical prints illustrating Anna Atkins' work. At left, a cyanotype photogram of the woodland horsetail (*Equisetum sylvaticum*), made by Anna Atkins and Anne Dixon in 1853, from their book, *Cyanotypes of British and Foreign Ferns.* [Image from Wikipedia, *http://en.wikipedia.org/wiki/Anna_Atkins* (accessed Feb 2008).] At right, a photogram of Algae, made by Anna Atkins as part of her 1843 book, *British Algae,* the first book composed entirely of photographic images. [Image from Wikimedia Commons, *http://en.wikipedia.org/wiki/Image:Anna_Atkins_algae_cyanotype.jpg* (accessed Feb 2008).]

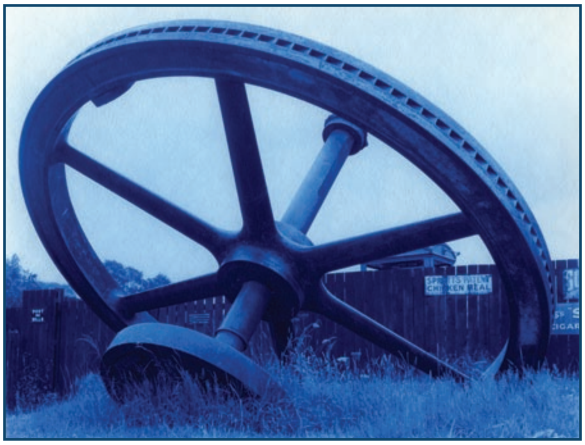
The Effects of Toning
A Selection of Mike Ware's Photographs

Corrugated barn, Brassington, Derbyshire. (Toned with tannic acid.)

"Snow Joke", Didsbury, Manchester. (Toned with lead(II) acetate.)

Corrugated window, Stanton in the Peak, Derbyshire. (Toned with gallic acid.)

"Stele", Scapa Flow, Orkney Islands. (Toned with trisodium phosphate.)



Wheel, Blist's Hill, Ironbridge, Shropshire. (Untoned cyanotype.)

Woodshop, Ironbridge, Shropshire. (Toned with thallium(I) sulfate.)

Figure 7. Cyanotype prints showing the effects of different tonings on Prussian blue. See also ref *24*. [All photographs courtesy of the author.]

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Winch Gears, South Ronaldsay, Orkney Islands. (Toned with tannic acid.) [Photograph courtesy of the author.]

to traditional photographic processes, as previously described in this *Journal (23)*.

The blue color of cyanotype is not inevitable: by applying a little chemical ingenuity, it is possible to transform it to other colors. The most widely-employed toning procedures for cyanotypes are those that first hydrolyze the Prussian blue, more or less completely, to iron(III) hydroxide, using a common alkali such as sodium carbonate solution or ammonia. The iron(III) hydroxide is then reacted with a chromogenic ligand such as gallic acid (3,4,5-trihydroxybenzoic acid) or tannic acid. The resulting blue–black (iron/gallate) or purplish–brown (iron/tannate) complexes provide satisfying image colors, and are the same constituents of the iron–gall inks that have been used as writing fluids since medieval times. If trisodium phosphate, Na_3PO_4, is used as the hydrolyzing alkali, then a bright yellow image in stable ferric phosphate, $FePO_4$, immediately results *(24)*. A selection of my photographs, showing different tonings, appears in Figure 7, opposite.

There are also more subtle ways of "re-tuning" the cyanotype blue. Some of the same metals that Prussian blue "mops up" so effectively, by trapping their cations in its lattice, can have a toning effect on cyanotypes. Thallium(I) turns it to a beautiful cornflower blue, but this cannot be recommended as a general practice, owing to the toxicity. Lead(II), in the form of a solution of lead(II) acetate, can also tone the pigment to a rich violet-blue by incorporation in the lattice; the sensitivity of this process to pH suggests that the lead(II) may be incorporated as $[Pb(OH)]^+$. Nickel(II) also has a toning effect, toward a greenish-blue, and has the added advantage of protecting the Prussian blue against alkaline hydrolysis. And so we find a union of the two disparate properties of Prussian blue that opened this essay: its zeolytic behavior to metal cations can be used to "tune" its electron charge–transfer spectrum, and so provide photographic artists with an enhanced palette of color for this most interesting of pigments.

Notes

1. The word "oxymoron" is deceptive-looking to a chemist; it means a figure of speech in which apparently contradictory terms appear together.
2. The name "Prussian Blue" has recently been appropriated by a young duo of pop singers—whose fascist lyrics prove to be far from "politically correct".
3. For clarity and simplicity in the equations quoted here, the formula for Prussian blue is taken as the stoicheiometric SPB, and the cation, which can vary, is generally omitted. It should be realized that solid Prussian blue can have various cations present, or none at all, and may not have this ideal formulation, but may tend towards the defect structure of IPB.

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Abstract and keywords

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Supplement

Video of the formation of Prussian blue

Dynamic Jmol structures of Prussian blue

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Classroom Activity Extension

Using "Blueprint Photography by the Cyanotype Process"

Related *JCE* Classroom Activity

Do you want to try the cyanotype process with your students? That's easy to do! Start with *JCE* Classroom Activity #19, "Blueprint Photography by the Cyanotype Process", by Glen D. Lawrence and Stuart Fishelson (*JCE,* **1999,** *76,* 1216A–1216B). In this ready-to-use activity, students create their own cyanotype paper and use it to make blueprint photographs in the sunlight. It's a great way to connect chemistry with art.

Others have also extended this technique. For example:

- Create cyanotypes on fabric using plant materials.
  Information about making cyanotypes on fabric and then making a quilt with these fabrics may be found online. See Making a Cyanotype Quilt—From the Garden at *http://www.alternativephotography.com/articles/art044.html* (accessed Feb 2008).
- Tone cyanotypes different colors.
  Find directions for toning cyanotypes—brown, violet, or split toning for a blue/yellow effect. Go to *http://www.ehow.com/how_2071359_tone-cyanotype.html* (accessed Feb 2008).

Ideas? Experiences? Tell Us!

If you have extended this activity in other ways, pass your ideas and experiences along to the *Journal* to share with other teachers. Contact Erica Jacobsen, *JCE* Secondary School Editor, at *jacobsen@chem.wisc.edu*.

Supporting *JCE Online* Material

http://www.jce.divched.org/Journal/Issues/2008/May/abs620.html

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